

AI Use Cases by Module

A detailed catalogue of AI/ML use cases across the EnPharChem platform

15 categories | 118 modules | 12 AI techniques | Generated May 12, 2026

Executive Summary

EnPharChem embeds artificial intelligence and machine learning into every module category - from seismic interpretation and reservoir simulation, through process simulation, exchanger design, FEED, MES, APC, supply chain, asset performance management, to grid management. This document tabulates the principal AI/ML use cases for each module, identifies the underlying technique, and states the expected business value, so that engineers, operators, and trainees can immediately recognise the AI surface area of every tool they use.

This catalogue is also a training reference: each row maps a module to the AI technique a learner should understand in order to operate that module safely and to extract value from it. Use it alongside the module-specific training courses available under Training > Courses.



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1. Process Simulation for Energy

18 modules

AI/ML augments first-principles thermodynamics with data-driven surrogates, accelerates convergence, predicts safety scenarios, and tunes refining reactor kinetics in real time.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem HYSYS	Surrogate Neural Networks; Bayesian Calibration	ML-augmented thermodynamic property prediction; AI-accelerated flowsheet convergence; data-driven tuning of EOS parameters.	Up to 70% faster steady-state convergence; reduced model re-tuning effort.
Acid Gas Cleaning	Gradient-Boosted Trees; LSTM	Predicts amine solvent loading, degradation rates, and CO2/H2S slip; flags abnormal absorber performance.	Lower solvent make-up costs; reduced corrosion incidents.
EnPharChem HYSYS Crude	Deep Neural Networks (TBP)	AI-based crude assay characterization; auto-generation of TBP and property curves from limited lab data.	Faster crude qualification; better refinery yield predictions.
BLOWDOWN Technology	Classification ML; Monte Carlo + ML	Risk classification of depressurization scenarios; auto-identification of low-temperature embrittlement risks.	Higher PSM compliance; fewer over-conservative designs.
Relief Sizing (Energy)	Anomaly Detection; Decision Trees	AI scenario generator for PSV sizing; detects unrealistic loads and missing scenarios in HAZOP coverage.	API 521/520 compliance with audit-ready trace.
Sulsim Sulfur Recovery	Reinforcement Learning; NN Surrogates	Optimizes Claus reactor air ratio; learns optimal tail-gas treatment trajectories.	>99.9% sulfur recovery; lower SO2 emissions.
Activated Economics (Energy)	LSTM Time-Series; Cost Indices ML	Forecasts equipment/material cost escalation; AI-driven CAPEX/OPEX scenario analysis.	More accurate Class 3/4 estimates.
Activated Energy Analysis (Energy)	Graph NN; AI Pinch Designer	AI proposes optimal heat-exchanger networks; predicts hot/cold composite curves from streams.	10-30% energy reduction via better integration.
Activated EDR (Energy)	NN Exchanger Surrogates	Replaces inner exchanger rating loops with fast NN surrogates inside flowsheet.	5-10x faster HX-coupled simulations.
EnPharChem HYSYS Petroleum Refining	Reinforcement Learning; Kinetic ML	Tunes FCC, reformer, hydrocracker kinetics to live yields; closes the loop with plant data.	Higher margin via yield optimization.
Refinery Reactor Models	RL Control; Surrogate NN	AI agent recommends WHSV, temperature, and H2/HC ratios.	Extended catalyst run length.
EnPharChem HYSYS Upstream	ML Flow Assurance; XGBoost	Predicts hydrate, wax, slugging; AI well production decline forecasting.	Reduced production downtime.
EnPharChem HYSYS Dynamics	Recurrent NN Surrogates	LSTM dynamic-process surrogates for fast transient studies and OTS.	Real-time-capable OTS scenarios.
Activated Dynamics	AI Operator Advisor (LLM-style)	Real-time guidance on abnormal-event response; transient anomaly classification.	Faster recovery from upsets.
EnPharChem Hybrid Models (Energy)	Hybrid First-Principles + ML	Native AI/ML hybrid: combines mechanistic skeleton with neural residual.	Industrial-grade extrapolation with data.
EnPharChem Multi Case (Energy)	Bayesian Optimization; AI DOE	Active learning across parametric studies; recommends next high-information case.	Fewer simulation runs for same insight.
EnPharChem Operator Training	LLM Coach; Scenario Generator	AI generates plausible upset scenarios; LLM answers operator questions in context.	Higher trainee competency, lower instructor load.
EnPharChem Flare System Analyzer (Energy)	CFD Surrogates; Emissions ML	NN-based flare radiation/noise contour prediction; AI emissions inventory.	Faster siting and permit compliance.

2. Process Simulation for Chemicals

18 modules

AI accelerates chemical process design through molecular-property neural networks, batch trajectory optimization, and automated documentation via LLMs.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Adsorption	Gaussian Process; NN Isotherms	AI fits Langmuir/Freundlich/Toth isotherms; optimizes PSA/TSA/VSA cycle times.	Higher productivity; lower energy per kg adsorbate.
EnPharChem Chromatography	CNN Peak Detection; AI Method DOE	Detects and deconvolutes peaks; AI gradient and mobile-phase optimization.	Reduced method development time.
EnPharChem Custom Modeler	Symbolic Regression	Equation discovery from data to extend custom unit models.	Captures novel equipment behavior.
EnPharChem Hybrid Models (Chemicals)	Hybrid Physics-ML; Auto-DOE	Combines mechanistic and data-driven layers; auto experimental design.	Faster scale-up; better generalization.
EnPharChem Multi Case (Chemicals)	Bayesian Sensitivity; Sobol-NN	AI sensitivity analysis prioritizes high-impact variables.	Sharper sensitivity insight per run.
EnPharChem Plus	Neural Surrogates; Convergence ML	NN surrogates for towers, reactors; learned convergence assistants.	2-10x simulation throughput.
EnPharChem Polymers	NN MWD Predictor	Predicts molecular-weight distribution from operating conditions.	Tighter product quality.
EnPharChem Properties	Graph Neural Networks (GNN)	Molecular GNN property prediction; auto thermo-model selection.	Faster property estimation for novel molecules.
Batch Modeling in EnPharChem Plus	RL Batch Trajectory; PAT-ML	Optimizes feed/temperature trajectories; integrates Process Analytical Technology.	Improved batch consistency and yield.
Distillation Modeling in EnPharChem Plus	NN Tray Hydraulics; AI Control	AI tray-loading prediction; ML soft sensors for product purity.	Higher recovery, less reboiler energy.
Relief Sizing (Chemicals)	Scenario ML; Anomaly Detection	AI scenario classifier for runaway-reaction PSVs.	DIERS-compliant sizing assistance.
Solids Modeling	Computer Vision; DEM Surrogate	CV-based particle-size distribution; NN surrogate for DEM solids simulation.	Faster solids design cycles.
Activated Economics (Chemicals)	LSTM Cost Forecasting	AI-driven economic evaluation under inflation scenarios.	Realistic NPV/IRR ranges.
Activated Energy Analysis (Chemicals)	Graph NN Heat Networks	AI proposes utility integration topology.	Lower utility cost.
Activated EDR (Chemicals)	NN Exchanger Surrogate	Embeds AI HX rating in chemical flowsheet.	Fast iteration on HX-bound designs.
EnPharChem Plus Dynamics	AI Controller Tuning	Auto-tunes PID/MPC for chemical dynamic models.	Better controllability scoring.
EnPharChem Process Manuals	LLM Document Generation	Auto-generates operating manuals, P&IDs narratives, and basis-of-design.	Cuts documentation time by 60-80%.
EnPharChem Flare System Analyzer (Chemicals)	CFD ML Surrogates	AI flare network rating with emissions inventory.	Faster compliance studies.

3. Exchanger Design & Rating

7 modules

AI replaces costly CFD/rigorous correlations with neural surrogates and predicts fouling/efficiency over operating life.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Air Cooled Exchanger	NN Fan Power; Weather ML	AI fan-speed optimization with ambient-temperature forecasting.	Lower aux-power consumption.
EnPharChem Fired Heater	NN Combustion Model; AI Efficiency	Predicts firebox temperature, NOx, and efficiency from fuel mix.	Lower fuel cost; reduced emissions.
EnPharChem Plate Exchanger	Fouling-Rate ML	Predicts plate fouling under variable services; recommends cleaning intervals.	Less unplanned shutdown.
EnPharChem Plate Fin Exchanger	NN Cryogenic Design	AI design assistant for brazed aluminum heat exchangers.	Optimized cryogenic LNG service.
EnPharChem Shell & Tube Exchanger	TEMA Design AI; Fouling ML	AI selects TEMA type; predicts fouling factors; auto pass-arrangement.	Faster shortlisting of viable designs.
EnPharChem Shell & Tube Mechanical	AI ASME Compliance Check	Auto stress/vibration checks; flags code non-conformance.	Reduced redesign iterations.
EnPharChem Coil Wound Exchanger	NN Cryogenic Thermal Surrogate	AI thermal model for LNG main exchangers.	Faster what-if studies.

4. Concurrent FEED

7 modules

Generative AI proposes 3D plant layouts; LLMs auto-draft engineering deliverables; ML forecasts cost escalation through FEED.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Fidelis	AI Cost-Process Linkage	Live link between simulation changes and CAPEX impact via ML cost models.	Faster concurrent FEED iterations.
EnPharChem Capital Cost Estimator	XGBoost; ML Location Factors	AI equipment-cost regression; ML location/labor factors.	Tighter Class 3 estimates.
EnPharChem In-Plant Cost Estimator	ML Revamp Forecasting	Predicts revamp cost from brownfield constraints.	Better turnaround budgeting.
EnPharChem Process Economic Analyzer	LSTM Economics; Scenario AI	AI NPV/IRR/payback forecasting under multiple price decks.	Risk-adjusted investment decisions.
EnPharChem OptiPlant 3D Layout	Generative AI Layout; Clash AI	GAN-based layout proposals; AI clash detection across disciplines.	Shorter FEED schedules.
EnPharChem OptiRouter	Reinforcement Learning Pipe Routing	RL agent learns rule-compliant pipe routes minimizing length and supports.	Faster, cheaper pipe routing.
EnPharChem Basic Engineering	LLM Auto P&ID; Datasheet AI	LLM drafts P&ID narratives, datasheets, and SOR.	Cuts basic engineering by 40-60%.

5. Subsurface Science & Engineering

11 modules

Deep learning transforms seismic processing, log interpretation, geological modelling, and reservoir simulation across the upstream lifecycle.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Subsurface Intelligence (ESI)	AI Data Fusion; ML Insights	Cross-domain subsurface fusion: seismic + well + production via ML.	Faster decision turnaround.
EnPharChem Echos	Deep Learning Denoise; CNN Velocity	AI seismic processing; CNN velocity-model building.	Cleaner images; less manual QC.
EnPharChem EarthStudy 360	Azimuthal Deep Learning	AI full-azimuth imaging and anisotropy detection.	Sharper reservoir characterization.
EnPharChem GeoDepth	Deep-Learning Migration	AI velocity update; learning-based RTM components.	Faster depth imaging.
EnPharChem SeisEarth	CNN Horizon/Fault Picking	Deep-learning auto-picking of horizons, faults, salt bodies.	70-90% interpretation time saved.
EnPharChem Geolog	ML Petrophysics; NN Log Synthesis	Predicts missing logs; AI lithology/facies classification.	Consistent petrophysical analysis.
EnPharChem RMS	AI Geostatistics; ML Property Modeling	NN-driven property modeling between wells.	Better history-match starting points.
EnPharChem SKUA	Deep-Learning Geomodels	AI structural framework building.	Faster geological model build.
EnPharChem OpsLink	Real-Time Anomaly Detection	AI monitors live field telemetry; detects equipment/well anomalies.	Reduced NPT (non-productive time).
EnPharChem Tempest	RL Well Control; ML History Matching	RL learns optimal well operating strategies; AI assisted history match.	Improved recovery factor.
EnPharChem Epos	AI Risk Scoring; Portfolio ML	AI exploration risk; ML portfolio optimization under uncertainty.	Better drilling capital allocation.

6. Energy & Utilities Optimization

3 modules

AI optimises site-wide energy, models decarbonization pathways, and forecasts utility demand for industrial sustainability.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Energy Analyzer	AI Pinch Designer; ML Forecasts	AI HEN synthesis; utility demand forecasting.	Site energy intensity reduction.
EnPharChem Sustainability Pathways	Optimization + ML; LCA AI	AI decarbonization-pathway optimizer; ML life-cycle inventory.	Net-zero roadmap with cost curves.
EnPharChem Utilities Planner	LSTM Demand Forecast; AI Scheduling	AI utility steam/power demand forecasting and scheduling.	Lower utility cost; reliability.

7. Operations Support

2 modules

Online AI keeps process simulation aligned with plant reality and provides intelligent assistants inside engineering workbooks.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem OnLine	Live ML Soft Sensors; Anomaly AI	Continuous data reconciliation + AI anomaly detection on live plant streams.	Closes simulation-to-plant gap.
EnPharChem Simulation Workbook	LLM Excel Assistant	AI recommends parameters, explains results, generates formulas inside workbook.	Faster engineering studies.

8. Advanced Process Control

7 modules

AI elevates DMC, nonlinear MPC, and soft sensors with automated tuning, deep learning models, and natural-language advisory.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem DMC3	AI MPC Tuning; ML Model Adaptation	Auto-tunes controller; adapts step models from plant data.	Sustained APC benefit capture.
EnPharChem Virtual Advisor (EVA) for DMC3	LLM Advisor	Natural-language Q&A on APC performance, root cause, and tuning.	Lower APC engineer load.
EnPharChem DMC3 Builder	AI System Identification	NN-based ARX/NARX identification; auto step-test design.	Faster controller commissioning.
EnPharChem Inferential Qualities	Deep-Learning Soft Sensors	CNN/LSTM soft sensors for online product quality.	Reduced lab dependency.
EnPharChem Nonlinear Controller	Neural NMPC	NN-based nonlinear MPC for strongly nonlinear units.	Tighter quality control.
EnPharChem Transition Management	Reinforcement Learning Transitions	RL learns optimal grade-changeover trajectories.	Less off-spec product.
EnPharChem Watch Performance Monitor	AI KPI Anomaly Detection	Detects controller degradation, valve issues, and CV drift.	Sustained APC service factor.

9. Dynamic Optimization

1 module

Closed-loop AI/RL optimization runs against live plant economics to capture real-time benefits.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem GDOT	Reinforcement Learning; NN Surrogates; RTO+ML	Real-time optimization with AI surrogates of process units; RL closed-loop economic optimization.	Continuous margin uplift across the unit.

10. Manufacturing Execution Systems

8 modules

AI enhances historian intelligence, batch deviation detection, and material balance reconciliation.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem InfoPlus.21	AI Tag Anomaly; ML Data Quality	Auto-flags drifting tags, frozen signals, and bad data quality.	Reliable downstream analytics.
EnPharChem Production Record Manager	AI Batch Deviation Detection	Detects parameter excursions in eBRs; auto-classifies deviations.	Faster QA release.
enPharChemONE Process Explorer	AI Trend Anomalies; LLM Search	Natural-language search of historian; AI anomaly highlights.	Faster troubleshooting.
EnPharChem Production Execution Manager	AI Workflow Optimization	AI recommends order sequencing and crew assignment.	Higher OEE.
EnPharChem Unified Reconciliation & Accounting	ML Reconciliation; AI Gross Error	Improves data reconciliation via ML weights; auto gross-error detection.	Closer mass balance closure.
EnPharChem Unified Movements	AI Tank Scheduling	Optimizes tank-to-tank movements with AI scheduler.	Reduced demurrage and contamination.
EnPharChem Operations Reconciliation	ML Yield Accounting	AI yield accounting and material balance closure.	Trustworthy operations KPIs.
EnPharChem Tank and Operations Manager	AI Tank-Level Forecasting	Predicts tank levels and ullage; AI alarming.	Avoids overflow/run-dry events.

11. Petroleum Supply Chain

11 modules

AI augments LP/NLP planning with warm-starts, validation, and LLM advisors that explain planning model results to schedulers.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Unified PIMS	AI Warm-Start; ML Model Validation	AI seeds LP/NLP solvers; ML detects unrealistic vectors.	Faster, more reliable plans.
EnPharChem Unified Scheduling	AI/RL Scheduler	RL learns near-optimal crude/product schedules.	Better vessel-tank-unit alignment.
EnPharChem Unified Multisite	AI Multi-Site Coordination	AI coordinates multi-refinery plans under shared constraints.	Network-level margin gain.
EnPharChem EVA for Unified PIMS	LLM Planning Advisor	Natural-language explanation of PIMS results and trade-offs.	Faster planner onboarding.
EnPharChem Verify for Planning	AI Model QA	ML detects stale yields, broken bounds, and structural issues.	Higher plan-vs-actual fidelity.
EnPharChem PIMS-AO	Advanced AI Optimization	AI successive-LP heuristics and integer-cut suggestions.	Better optimum on tough cases.
EnPharChem Assay Management	AI Assay Characterization	Auto-builds crude assays from sparse lab data.	Faster new-crude qualification.
EnPharChem Petroleum Scheduler	AI Operations Scheduler	Detailed scheduling with AI conflict resolution.	Reduced demurrage and giveaway.
EnPharChem Refinery Multi-Blend Optimizer	ML Blend Quality Prediction	NN blend property models for non-linear quality interactions.	Less giveaway, more spec compliance.
EnPharChem Petroleum Supply Chain Planner	AI E2E Planning	AI-supported crude-to-product chain planning under uncertainty.	Resilient supply chain plans.
EnPharChem Collaborative Demand Manager	LSTM Demand Forecasting	AI demand sensing for petroleum products.	Better S&OP accuracy.

12. Supply Chain Management

6 modules

AI/ML powers demand forecasting, multi-echelon inventory optimization, and KPI insight generation for chemical supply chains.

Module	AI / ML Technique	AI Use Case	Business Outcome
enPharChemONE Supply Chain Management	AI Integrated SCM	AI orchestration across plan/source/make/deliver.	End-to-end SCM resilience.
EnPharChem Scheduler Explorer	AI/RL Scheduling	RL-based plant scheduling with constraint learning.	Higher throughput and OTIF.
EnPharChem Collaborative Demand Manager (SCM)	LSTM/Prophet/Bayesian Demand	Hierarchical demand forecasting with collaboration.	Improved forecast accuracy.
EnPharChem Supply Chain Planner	AI Multi-Echelon Optimization	AI inventory and replenishment optimization.	Lower working capital.
EnPharChem Plant Scheduler Family	AI Scheduling	AI sequence-dependent scheduling.	Reduced changeover loss.
EnPharChem SCM Insights	AI KPI Anomaly; LLM Insights	Auto-narrated SCM insights and AI anomaly flags.	Faster managerial action.

13. Asset Performance Management

4 modules

Native AI/ML is the heart of APM: pattern recognition for failure prediction, multivariate analytics, and chemometric data analysis.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Mtell	ML Pattern Recognition (Failure Agents)	Learns failure signatures and predicts equipment breakdowns weeks ahead.	20-40% maintenance cost reduction.
EnPharChem ProMV	PCA/PLS Multivariate Analytics	Multivariate statistical process and batch monitoring with contribution plots.	Early quality and process deviation alarms.
EnPharChem Process Pulse	Real-Time Anomaly AI	Live anomaly detection across operating envelopes.	Prevented unplanned shutdowns.
EnPharChem Unscrambler	Chemometrics; Neural Networks; DOE	Multivariate calibration, classification, and design of experiments.	Faster R&D, better PAT models.

14. Industrial Data Fabric

1 module

AI gives industrial data context: semantic enrichment, NLP-based metadata, and ML data-quality scoring.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Inmation	NLP Metadata; AI Semantic Enrichment; ML Quality	AI auto-tags industrial data, builds contextual graphs, and scores data quality.	Trustworthy data foundation for all AI use cases.

15. Digital Grid Management

14 modules

AI underpins modern grid: forecasting, DER orchestration, outage prediction, FLISR/VVO automation, and storm resilience.

Module	AI / ML Technique	AI Use Case	Business Outcome
EnPharChem Microgrid Management System	RL Microgrid Control; AI EMS	RL agent optimizes microgrid dispatch with PV/wind/BESS forecasts.	Lower energy cost and higher reliability.
EnPharChem OSI Monarch SCADA	AI Alarm Management; ML Anomaly	AI alarm rationalization; ML SCADA anomaly detection.	Reduced operator overload.
EnPharChem OSI Generation Management	AI Dispatch; Renewables Forecast	AI economic dispatch with wind/solar forecasting.	Lower marginal generation cost.
EnPharChem OSI Energy Management	Deep-Learning State Estimation	AI EMS state estimation and contingency analysis.	Higher transmission reliability.
EnPharChem OSI Advanced Distribution Mgmt	AI FLISR; ML VVO; Outage AI	AI-driven fault location/isolation/service restoration; volt/var optimization.	Lower SAIDI/SAIFI.
EnPharChem OSI DER Management	AI DER Forecasting; RL Aggregation	AI forecasts and orchestrates distributed energy resources.	Higher DER hosting capacity.
EnPharChem OSI CHRONUS Historian	AI Grid Analytics	AI analytics on grid historian (event detection, classification).	Operational intelligence for grids.
EnPharChem OSI Continua Pipeline Management	ML Leak Detection; AI Batch Tracking	AI leak detection and batch tracking on pipelines.	Faster leak response, better integrity.
EnPharChem Cimphony Network Model Management	AI CIM Validation	AI validates CIM network models for consistency.	Cleaner model handoff to operations.
EnPharChem Grid Apps	ML DLR; AI Hosting Capacity; LSTM Load	Dynamic line rating, hosting-capacity ML, AI load forecasting.	More headroom on existing assets.
EnPharChem Grid Reporter	LLM Regulatory Reports	LLM-generated regulatory and operational reports.	Compliance with less effort.
EnPharChem DER Connect	AI DER Orchestration	AI DER interconnection and communications optimization.	Faster DER onboarding.
EnPharChem Network Maps	AI Geospatial Analytics	AI geospatial analytics across grid assets.	Smarter network planning.
EnPharChem Resilience Portal	AI Storm/Outage Prediction	AI storm-impact forecasting and outage prediction.	Higher grid resilience.

Cross-cutting AI / ML Technique Index

Technique	Description
Hybrid Physics-ML	Combines first-principles equations with neural-net residual learning. Used in HYSYS Hybrid Models, EnPharChem Plus Hybrid Models, APM, ESI.
Surrogate Neural Networks	Fast NN replacements for expensive rigorous unit operations (towers, exchangers, reactors).
LSTM / Recurrent Networks	Time-series forecasting (demand, equipment health, dynamic process transients, economics).
Reinforcement Learning	Sequential decision-making: dispatch, scheduling, grade transitions, RTO, well control, microgrid.
CNN / Deep Learning	Spatial pattern recognition: seismic interpretation, particle imaging, flare CFD surrogates.
Graph Neural Networks	Molecular property prediction; heat-exchanger network synthesis.
Multivariate Statistics (PCA/PLS)	Batch and continuous-process monitoring (ProMV, Unscrambler).
Anomaly Detection	Catches drift, gross errors, sensor failures, abnormal operation.
LLMs (Large Language Models)	Documentation generation, NL Q&A assistants, regulatory reports, EVA-style advisors.
Bayesian Optimization / Active Learning	Picks the next most-informative simulation case or experiment.
Symbolic Regression	Discovers analytical equations from data inside custom modelers.
Generative AI (GANs/Diffusion)	Generative plant layouts, scenario synthesis.